# Python Functions – Zoho/MNC Interview Pattern

20-Question Interview Set Focused on Pre-Defined & User-Defined Functions

**Total Questions:** 20 **General Rule:** Python only, No imports/modules

## 🟢 NUMBER LOGIC & FUNCTIONS (Questions 1 – 5)

### 1. Reverse a Number

Write a Python program using a user-defined function to reverse a given positive integer. The function should extract each digit from the number and construct the reversed number. You should not convert the number into a string.

**Input Format:** A single positive integer N. | **Output Format:** Print the reversed number. | **Constraints:** 1 <= N <= 10^9. Do not convert to string.

```
Sample Input:
12345

Sample Output:
54321
```

### 2. Sum of Digits

A billing system generates a numeric transaction ID. Write a user-defined function that calculates the sum of all digits in the transaction ID. For example, if the transaction ID is 5832, the sum is: 5 + 8 + 3 + 2 = 18.

**Input Format:** A single positive integer N. | **Output Format:** Print the sum of its digits. | **Constraints:** 1 <= N <= 10^9.

```
Sample Input:
5832

Sample Output:
18
```

### 3. Prime Number Checker

A company wants to check whether a given employee ID number is a prime number. Write a user-defined function is_prime() that accepts an integer and returns whether the number is prime. A prime number has exactly two factors: 1 and itself.

**Input Format:** A single integer N. | **Output Format:** Print "Prime" or "Not Prime". | **Constraints:** 1 <= N <= 10^6.

```
Sample Input:
29

Sample Output:
Prime
```

### 4. Prime Numbers in a Range

A recruitment application needs to identify all prime numbers within a given range. Write a user-defined function is_prime() and use it to find all prime numbers between L and R.

**Input Format:** Two integers L and R. | **Output Format:** Print all prime numbers between L and R separated by spaces. | **Constraints:** 1 <= L <= R <= 10^5.

```
Sample Input:
10 30

Sample Output:
11 13 17 19 23 29
```

### 5. Palindrome Number

A number is called a palindrome if it remains the same when reversed. Write a user-defined function is_palindrome() to determine whether the given number is a palindrome. You must solve this using numerical operations and should not convert the number into a string.

**Input Format:** A single positive integer N. | **Output Format:** Print "Palindrome" or "Not Palindrome".

```
Sample Input:
1221

Sample Output:
Palindrome
```

## 🟡 STRING-BASED FUNCTIONS (Questions 6 – 10)

### 6. Reverse a String

Write a user-defined function reverse_string() that accepts a string and returns the string in reverse order. Do not use the built-in reverse() function.

**Input Format:** A single string. | **Output Format:** Print the reversed string. | **Constraints:** Contains alphabets, digits, spaces. Do not use reverse().

```
Sample Input:
programming

Sample Output:
gnimmargorp
```

### 7. Count Vowels

A text-processing application needs to determine how many vowels are present in a given sentence. Write a user-defined function count_vowels() that counts: a, e, i, o, u in both uppercase and lowercase.

**Input Format:** A string containing a sentence. | **Output Format:** Print the total number of vowels.

```
Sample Input:
Python Programming

Sample Output:
4
```

### 8. Character Frequency

A company wants to analyze a user-entered message and determine how frequently each character occurs. Write a user-defined function that accepts a string and displays the frequency of every character. Characters should be counted separately.

**Input Format:** A single string. | **Output Format:** Display each character and its frequency.

```
Sample Input:
banana

Sample Output:
b : 1
a : 3
n : 2
```

### 9. First Non-Repeating Character

Given a string, find the first character that appears exactly once. For example, in "aabbcdde", 'c' appears only once and is the first non-repeating character. Write a user-defined function to solve the problem.

**Input Format:** A single string. | **Output Format:** Print the first non-repeating character, or "No unique character".

```
Sample Input:
aabbcdde

Sample Output:
c
```

### 10. Remove Duplicate Characters

Write a user-defined function remove_duplicates() that removes repeated characters from a string. The first occurrence of each character must be retained, and the original order must be maintained.

**Input Format:** A single string. | **Output Format:** Print the string after removing duplicate characters.

```
Sample Input:
programming

Sample Output:
progamin
```

## 🔵 LIST + FUNCTION QUESTIONS (Questions 11 – 15)

### 11. Find the Second Largest Number

An organization stores employee performance scores in a list. Write a user-defined function to find the second-largest distinct number. You must not use sort(), sorted(), or max().

**Input Format:** A list of integers. | **Output Format:** Print the second-largest number.

```
Sample Input:
10 45 23 67 34

Sample Output:
45
```

### 12. Remove Duplicate Elements from a List

A database contains duplicate customer IDs. Write a user-defined function to remove duplicate IDs while maintaining the original order.

**Input Format:** A list of integers. | **Output Format:** Print the list after removing duplicates.

```
Sample Input:
10 20 10 30 20 40

Sample Output:
10 20 30 40
```

### 13. Move All Zeros to the End

A data-processing system contains a list where 0 represents missing data. Move all zeros to the end of the list while keeping the order of all non-zero elements unchanged. Write a user-defined function move_zeros().

**Input Format:** A list of integers. | **Output Format:** Print the modified list.

```
Sample Input:
0 5 0 3 8 0 2

Sample Output:
5 3 8 2 0 0 0
```

### 14. Find the Missing Number

A system stores numbers from 1 to N. Due to a data-entry error, exactly one number is missing. Write a user-defined function to find the missing number.

**Input Format:** First line: N, Second line: N-1 integers. | **Output Format:** Print missing number. | **Constraints:** 1 <= N <= 10^5.

```
Sample Input:
6
1 2 3 5 6

Sample Output:
4
```

### 15. Find Pairs With Given Sum

A company analyzes transaction amounts and wants to identify pairs whose sum equals a given target value. Write a user-defined function that finds all such pairs. Each pair should contain two different elements from the list.

**Input Format:** First line: numbers, Second line: target value. | **Output Format:** Print matching pairs.

```
Sample Input:
2 7 11 15 3 6
9

Sample Output:
2 7
3 6
```

## 🔴 ADVANCED ZOHO-STYLE LOGIC QUESTIONS (Questions 16 – 20)

### 16. Anagram Without Sorting

Two strings are considered anagrams if they contain the same characters with the same frequencies. Write a user-defined function is_anagram() to check whether two strings are anagrams. You must not use sort() or sorted().

**Input Format:** Two strings. | **Output Format:** Print "Anagram" or "Not Anagram".

```
Sample Input:
listen
silent

Sample Output:
Anagram
```

### 17. String Compression

A text-processing system wants to reduce the size of repeated consecutive characters. Write a user-defined function that compresses consecutive characters by storing the character followed by its count. (e.g., "aaabbcccc" becomes "a3b2c4").

**Input Format:** A string containing lowercase characters. | **Output Format:** Print the compressed string.

```
Sample Input:
aaabbcccc

Sample Output:
a3b2c4
```

### 18. Find Leaders in an Array

In an array, an element is called a leader if it is greater than every element appearing to its right. The rightmost element is always considered a leader. Write a user-defined function find_leaders() to identify all leader elements.

**Input Format:** A list of integers. | **Output Format:** Print all leader elements from left to right.

```
Sample Input:
16 17 4 3 5 2

Sample Output:
17 5 2
```

### 19. Employee Salary Ranking

A company stores employee names and salaries. Write a user-defined function to determine the employee with the highest salary and the employee with the second-highest salary. Do not directly use max() or sorted().

**Input Format:** Employee names and salaries. | **Output Format:** Highest and Second Highest employee records.

```
Sample Input:
Arun 45000
Priya 52000
Kumar 48000
Divya 60000

Sample Output:
Highest: Divya 60000
Second Highest: Priya 52000
```

### 20. Student Result Processing ⭐

A college wants to automate student result processing. Write a user-defined function calculate_result() that accepts marks in five subjects. Calculate total, average, grade (90-100: A, 80-89: B, 70-79: C, 60-69: D, <60: F), and pass/fail (Pass if average >= 40).

**Input Format:** Five integer marks separated by spaces. | **Constraints:** Marks between 0 and 100.

```
Sample Input:
78 85 67 92 88

Sample Output:
Total: 410
Average: 82.0
Grade: B
Result: Pass
```